

# Thomas Chisica Londoño

Applied Mathematics & Computer Science

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## RESEARCH INTERESTS

Reinforcement learning, computational neuroscience, and reliable AI, including LLM-based systems. My work combines mathematical modelling, simulation, and controlled experiments.

## EDUCATION

- 2024–present BSc in Applied Mathematics and Computer Science at **Universidad del Rosario**, Bogotá. GPA: 4.11/5.00. Expected graduation: June 2028.
- Coursework Machine Learning (5.0/5.0), Optimisation, Algorithm Analysis & Design, Operating Systems, Linear Algebra, Probability & Statistics I–II, Real Analysis, Graph Theory, Differential Equations, Numerical Analysis & Scientific Computing, Theory of Computation.

## RESEARCH EXPERIENCE

### Evaluating reinforcement learning for supply-chain control 2025–present

*Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario*

- Reconstructed a thirteen-operation supply chain in SimPy and built Gymnasium environments and PyTorch/PPO experiments to test whether learned control improves resilience under disruptions.
- Designed a comparison giving static policies the same decision options as PPO, with separate calibration and held-out simulation runs. Across 60 test runs and 10 PPO checkpoints, the stronger static comparator overturned the apparent learning advantage.
- Tested retained-versus-reset online adaptation across disruption cycles; maintained automated tests, source-model checks, and reproducible experiment records.

 [SCRES: code & experiments](#)

### Collective behaviour and resource allocation 2025–present

*Prof. Edgar Andrade-Lotero, Universidad del Rosario; UC Davis collaboration*

- Contribute to the team’s Foragers research on how resource-sharing rules and fairness relate to collective outcomes, including group survival under scarcity.
- Implemented the Gymnasium simulation’s investment, allocation, harvest, and redistribution pipeline, with a wealth-minus-inequality reward. Added tests of rewards, state bounds, termination, and full simulation episodes.

### Auditing models of marmot social networks 2026–present

*Research collaboration with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario*

- Audited a legacy NetLogo/R model of social homophily. Traced agent–string comparisons that collapsed distinct model conditions, using a Python reference implementation and native NetLogo 5.2 smoke tests across 240 model-condition runs.
- Built provisional dyadic analysis tables linking interactions, eligibility, and observation effort, with checks for exposure and zero-event cases. Documented why annual group membership alone cannot establish simultaneous availability or partner preference.

## RESEARCH PROJECTS & TRAINING

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### Spectral analysis of human collective search

Independent project

- Identified a degenerate Fiedler eigenspace on the  $8 \times 8$  board in public collective-search data; compared human partitions with symmetry-aware references.
- Tested early predictors using leave-one-out cross-validation on 29 dyads (exploratory AUC: 0.804 geometric; 0.860 combined). Released reproducible code, tables, figures, and limitations.

 [Spectral analysis](#)

### Neuromatch Academy: Computational Neuroscience

July 2026

*68-hour interactive course; team project and independent follow-up*

- Studied connectivity in recurrent networks for a motor task, comparing four densities with eight paired seeds and held-out evaluation in the primary experiment.
- Identified that density also changed the number of trainable recurrent weights. Ran a follow-up holding that number fixed; the original density trend did not recur under this control.

 [Motor RNN: experiments & controls](#)

## PROFESSIONAL EXPERIENCE

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### Lead in Emerging Technologies

June–December 2025

*Digital Business Ventures; assigned to Witty with Progresión*

- Built a pipeline to process customer-support emails and recurring questions into a knowledge base for Witty, a financial-services platform used by Progresión clients.
- Developed an LLM-based support chatbot using FastAPI, FAISS and multi-provider embeddings, with heuristic answer-or-escalate routing. Designed it to handle routine queries at low inference cost.

## PROJECTS

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### HeliOS: educational RISC-V kernel

 [GitHub](#)

Contributed to a collaborative C kernel with timer-driven scheduling, a UART shell, synchronisation primitives, and heap allocation. The repository includes QEMU/OpenSBI smoke tests.

### ChaosLab: double-pendulum simulation

 [GitHub](#)

Built a Python/Streamlit double-pendulum simulation with numerical integration, energy-conservation checks, phase-space trajectories, and flip-time maps.

## TEACHING, AWARDS & FUNDING

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|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024–2025    | <b>Calculus I Teaching Assistant</b> , Universidad del Rosario (July 2024–July 2025). Led problem-solving sessions and supported first-year students. |
| 2024         | <b>Andrés Bello National Distinction</b> . Ranked 33rd nationally in Colombia's 2023 secondary-school examination (ICFES Saber 11).                   |
| 2024–present | <b>Merit scholarship covering 100% of tuition</b> , Universidad del Rosario.                                                                          |

## SKILLS

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|----------------|-----------------------------------------------------------------|
| Programming    | Python, C, R, SQL.                                              |
| ML & modelling | PyTorch, Gymnasium, SimPy, NumPy/SciPy, NetworkX.               |
| Development    | Git, Linux, pytest, continuous integration, FastAPI, $\LaTeX$ . |
| Languages      | Spanish (native); English (C1, IELTS, 2023).                    |